

DiagBoost: Further Optimizing SpinQuant via Affine Transformations

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Abstract—Low-bit quantization is crucial for reducing model size and improving inference efficiency. However, existing post-training quantization methods deal with accuracy degradation caused by outliers in weight and activation distributions. We propose DiagBoost, a generalized extension of SpinQuant that augments learned rotations with a bounded, learned diagonal matrix. This diagonal scaling increases flexibility while adding minimal computational overhead since diagonal operations are cheap, hardware-efficient, and require few parameters.

We evaluated DiagBoost on Isotropic, Diagonal, and SPD datasets, as well as on GPT-2 trained on WikiText-2. On anisotropic data, DiagBoost reduced reconstruction error by 15-60% compared to baseline 4-bit quantization and SpinQuant while also matching performance on isotropic data. On GPT-2, DiagBoost significantly improved perplexity and loss. Based on these results, DiagBoost has strong potential for improving low-bit quantization in modern transformer models.

Index Terms—quantization, affine transformations, rotations

I. INTRODUCTION

Modern neural networks require significant memory and computational resources, motivating the use of low-bit quantization to improve inference efficiency. However, aggressive quantization can harm accuracy due to the presence of outliers in weight matrices that expand the dynamic range and force larger quantization step sizes. Although clipping these outliers may reduce this error, we simultaneously lose important information, thus establishing a fundamental challenge in the field of quantization.

Prior work has attempted to tackle these issues, such as SmoothQuant, which rescales activations with diagonal matrices, and SpinQuant, which applies learned orthogonal rotations to reduce kurtosis before quantization [1] [2]. However either diagonal scaling or pure rotations on their own are not enough to handle these issues. More expressive affine transformations are possible but can introduce excessive computational costs, making them impractical for real-world deployment.

To address these challenges, we propose DiagBoost, a generalization of SpinQuant that applies a learned rotation followed by a bounded diagonal matrix. This diagonal component provides additional scaling flexibility while adding minimal parameters. An overview of the architecture that we have implemented is available in Figure 1.

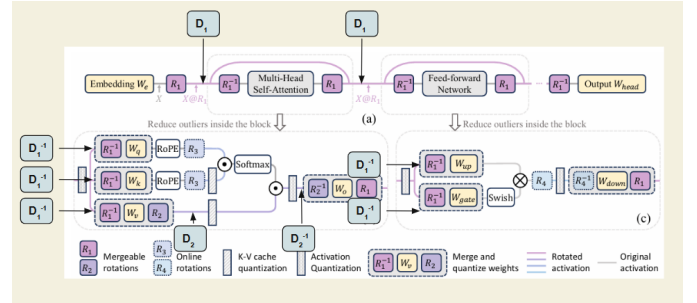


Fig. 1. Diagram of DiagBoost architecture.

In addition, we applied DiagBoost to a real transformer model (GPT-2) trained on WikiText-2, where weight MSE remains stable across methods but loss and perplexity showed substantial improvement.

II. APPROACH

A. Intuition

Quantization error depends heavily on the shape of weight or activation distributions. For example, some real-world neural network activations form elongated ellipses, which results in a few outlier-heavy dimensions dominating the dynamic range. Because uniform quantization has to cover this full range, these outliers force larger step sizes and increase rounding error.

SpinQuant mitigates this by applying a learned orthogonal rotation that reorients the distribution so that its variance is more evenly distributed across axes. This aligns the ellipse with the coordinate system, which reduces kurtosis and makes uniform quantization more effective [1].

However, rotation on its own can't change the length of the axes, just the orientation. DiagBoost extends this intuition by introducing a learned but bounded diagonal matrix that rescales each axis independently, further creating a more spherical distribution. More formally, our bounding relies on a parameter ρ , such that each dimension of the activations are scaled by $\lambda \in [e^{-\rho}, e^{\rho}]$. Thus, by lowering ρ , we can ensure that the norm of the matrix $A = RD$ remains close to 1. In

the best-case scenario, we significantly improve quantization, and in the worst-case, the diagonal reduces to the identity, reverting to standard SpinQuant with minimal overhead.

The power of these bounded diagonal matrices cannot be underestimated: not only do they provide more degrees of freedom (and thus potential for superior results) at the cost of relatively few extra parameters (enabling fast training), they are also structured, enabling hardware optimization.

B. DiagBoost Architecture

DiagBoost generalizes SpinQuant by adding a bounded, learnable diagonal matrix immediately after the learned rotation. SpinQuant applies a transformation $Y = RX$, where R is an orthogonal rotation matrix that reduces kurtosis; with DiagBoost, this is extended to the affine transformation

$$Y = AX = RDX$$

, where D is a constrained diagonal matrix that provides coordinate-wise flexibility/scaling while R maintains norm preservation. After quantization, the original representation is recovered using $A^{-1} = D^{-1}R^\top$.

Because diagonal operations cost only $O(d)$ and are highly hardware-efficient, DiagBoost minimizes overhead. In addition, it supports joint training, where R and D are optimized together, and separate training, where R is learned first and then frozen while D is optimized.

C. Mathematical Proof-of-Concept

We claim that DiagBoost theoretically achieves a strictly lower quantization error than SpinQuant for any anisotropic dataset, with equality on isotropic datasets. Consider the following expected reconstruction loss:

$$\mathcal{E}(A) = E\|X - A^{-1}Q(AX)\|_2^2$$

for a d -dimensional square matrix X and affine transform A , where Q is a simulated b -bit symmetric quantization function. For the rest of this section, we assume 4-bit quantization ($b = 4$), but the same result holds for different b . Writing the quantization error as:

$$Q(Y) = Y + \varepsilon(Y) = Y + \varepsilon_{\text{round}}(Y) + \varepsilon_{\text{clamp}}(Y)$$

we can control clamping by selecting the quantization half-range α , such that we effectively only focus on $\varepsilon_{\text{round}}(Y)$. Thus we are really trying to minimize:

$$\mathcal{E}(A) \approx E\|A^{-1}\varepsilon_{\text{round}}(Y)\|_2^2$$

The quantization function Q is defined as:

$$Q(y) = \Delta \cdot \text{clamp}\left(\text{round}\left(\frac{y}{\Delta}\right), -r, r\right)$$

and $r = 2^{b-1} - 1 = 7$, $\Delta = \frac{2\alpha}{2^b-1} = \frac{2\alpha}{15}$. To minimize clamping, we may select α such that the probability of clipping falls below some predefined range, such as 0.1%: this can be trivially done by setting it to the max value, but at the cost of larger rounding error.

It can be shown that if we want $P(\text{clamping occurs}) < d$ for some d , then if X is an activation vector, for fixed $Z = DX$ and a random Haar matrix R :

$$\alpha(X; D) = \sqrt{\frac{2 \log(2d/\delta)}{d}} \|DX\|_2$$

To minimize reconstruction error, we may expand like so:

$$\begin{aligned} E[\|A^{-1}\varepsilon_{\text{round}}(DX)\|_2^2 | X] &= \frac{\Delta(X; D)^2}{12} \text{tr}(A^{-\top} A^{-1}) \\ &= C_{4\text{-bit}} \cdot \frac{2 \log(2d/\delta)}{d} \text{tr}(D^2 \Sigma) \cdot \text{tr}(D^{-2}) \end{aligned}$$

where $C_{4\text{-bit}} = 0.002963$ is a constant coming from the fact that $b = 4$. We then aim to minimize:

$$J(D) = \text{tr}(D^2 \Sigma) \cdot \text{tr}(D^{-2})$$

Note that R has completely disappeared. Thus, if we can prove that the optimal diagonal matrix D^* is not the identity I , then our method has a provable improvement over SpinQuant, as SpinQuant is simply $A = RD$ with $D = I$.

Diagonalizing $\Sigma = U\Lambda U^\top$ with $\Lambda = \text{diag}(\lambda_i)$, then if $D = \text{diag}(d_i)$, we obtain:

$$J(D) = \left(\sum_{i=1}^d d_i^2 \lambda_i \right) \left(\sum_{i=1}^d \frac{1}{d_i^2} \right)$$

Then by the Cauchy-Schwartz inequality, the optimal diagonal is $D^* = \text{diag}(\lambda_i^{-1/4})$ up to a global scale. We then have:

$$J(D^*) = \left(\sum_{i=1}^d \sqrt{\lambda_i} \right)^2 \leq d \sum_{i=1}^d \lambda_i = J(I)$$

again by Cauchy-Schwartz, with strict inequality unless $\lambda_1 = \dots = \lambda_d$ (isotropic activations), which does not occur in real-world datasets. Thus, there exists an optimal diagonal D^* to train towards, which is strictly superior to $D = I$.

III. IMPLEMENTATION

We implemented DiagBoost in Google Colab, using A100 GPUs for training. We injected learnable rotation and diagonal matrices before the self-attention layers and the MLPs of a pretrained/base GPT-2 Small (124M parameters), then trained SpinQuant and DiagBoost: this was done by creating LearnableRotation and LearnableDiagonal modules, each containing a trainable rotation/diagonal matrix respectively. Quantization was simulated using a SimQuantLinear module, with back-propagation done with STE. To increase the training speed of rotation matrices, these matrices were optimized via the Cayley transform of an internal skew-symmetric matrix.

DiagBoost uses two different training regimes: joint training (training both rotation and diagonal matrices for N epochs), and separate training (training only rotations for $N/2$ epochs, with $D = I$, then freezing rotation matrices and training exclusively diagonal matrices for $N/2$ epochs).

We simulated 4-bit symmetric quantization on weights and 8-bit symmetric quantization on activations, due to the higher variance of activations. We used `batch_size = 8` and

a block size of 256 tokens. We trained on the Wikitext-2 dataset, specifically wikitext-2-raw-v1, which is more difficult and more commonly used for modern research. We used 2 GPU workers, and set our diagonal bounding hyperparameter $\rho = 0.5$. We used an AdamW optimizer with learning rates of 10^{-4} and 10^{-5} (separate tests), and no warmup was used. We trained for $N = 20$ epochs for SpinQuant and both DiagBoost training regimes, with training time measured via wall-clock time. The loss used was cross-entropy loss, and perplexity was derived as e^{loss} .

IV. RESULTS

We first evaluated DiagBoost on GPT-2 (small) trained on WikiText-2-raw. Table I reports results from our initial experiments, where we trained with learning rate 10^{-4} and diagonal bound $\rho = 2$. All methods use 4-bit symmetric weight quantization and 8-bit symmetric activation quantization.

TABLE I
GPT-2 QUANTIZATION RESULTS ON WIKITEXT-2 (INITIAL SETTING, LR = 10^{-4} , $\rho = 2$).

Method	Loss	Perplexity
Baseline	4.261799	70.937508
SpinQuant	4.228531	68.616386
DiagBoost (joint)	4.106576	60.738361

As shown in Table I, both SpinQuant and DiagBoost substantially improve language modeling performance over the baseline quantization. DiagBoost (joint training of R and D) achieves the lowest loss, improving perplexity by roughly 10 points compared to the baseline and by almost 8 points compared to SpinQuant, despite introducing only a diagonal scaling.

However, we observed that aggressively training DiagBoost to minimize loss came at the cost of slightly worse activation quantization error. To better balance model performance and activation fidelity, we reduced the learning rate to 10^{-5} and tightened the diagonal bound from $\rho = 2.0$ to $\rho = 0.5$. We also evaluated a separate-training regime where we first train only the rotations and then freeze them while training the diagonals. The corresponding results are summarized in Table II.

TABLE II
EFFECT OF CONSERVATIVE HYPERPARAMETERS ON ACTIVATION ERROR, PERPLEXITY, AND TRAINING TIME (LR = 10^{-5} , $\rho = 0.5$).

Method	Act. MSE	Perplexity	Time (s)
Baseline	0.012195	70.937508	—
SpinQuant	0.012339	64.855614	8950.13
DiagBoost (joint)	0.013010	65.001129	9144.85
DiagBoost (separate)	0.012807	65.166138	8255.19

Under these more conservative hyperparameters, all methods exhibit very similar activation reconstruction error, with DiagBoost incurring only a small increase in activation MSE relative to the baseline. SpinQuant achieves the best perplexity in this setting, while both DiagBoost regimes remain competitive and still provide a large improvement over the

baseline. Importantly, the separate-training variant of DiagBoost achieves nearly the same perplexity as joint training, but reduces wall-clock training time by approximately 10% compared to SpinQuant. This makes sense, as DiagBoost (separate) trains rotation matrices for only half the epochs that SpinQuant does, and even with optimized training routines for rotations, training diagonals is still far faster because of the low number of parameters.

Overall, our experiments indicate that augmenting SpinQuant with a bounded diagonal matrix can significantly improve model perplexity at fixed weight MSE, and that with appropriate hyperparameters and training regimes, DiagBoost attains a favorable trade-off between activation error, language modeling performance, and training efficiency.

V. RELATED WORK

Our method is built upon and is a generalization of SpinQuant, which injects learned rotation matrices into the attention and MLP layers of transformer-based architectures. This greatly reduces kurtosis of activations, allowing them to be more readily quantized [1].

AffineQuant also utilizes affine transformations to aid quantization, but AffineQuant precedes SpinQuant and uses strictly diagonal dominant matrices instead of rotations [3]. SmoothQuant also introduces diagonals to aid quantization, but lacks rotation matrices [1]. These methods primarily focus on decreasing perplexity and loss, while our method also focuses on training time efficiency. Additionally, SmoothQuant lacks a rigorous justification of why their method works, relying on empirical results: we present a proof that our method yields decreased quantization error over SpinQuant [2].

VI. CONCLUSION

On GPT-2, for higher learning rates, DiagBoost achieves a significantly lower perplexity and loss than SpinQuant. For lower learning rates, DiagBoost using the separate training regime can achieve significantly shorter training times than SpinQuant by training rotation matrices for fewer epochs, with near-equal results in perplexity and quantization error.

In the future, we hope to test DiagBoost on larger models such as Llama 3.2-1B to see if the same improvements hold. In addition, it would be interesting to consider whether the benefits of DiagBoost also apply to vision models, such as ViT or CNN-based models, particularly because image-based datasets significantly differ in distribution from text-based datasets. Theoretically, for any real-world representations of data with strong anisotropy (including KV caches, embedding vectors, gradient updates, and covariance-like features), DiagBoost achieves strictly lower quantization error, but because there is (as of yet) no established correlation between quantization error and loss, there is further work to be done on whether these theoretical results on quantization error also translate to loss reduction.

GROUP CONTRIBUTION STATEMENT

Sam wrote the proof and training code. Advikar wrote result extraction code. Josh and Nathan researched literature, with Nathan coming up with separate R, D training. All worked on the report.

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